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Density functional study on the mechanics, thermodynamics, and H diffusion mechanism of LiH

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HIGHLIGHTS

- First-principles calculations can allow for a better understanding of the H diffusion mechanism in LiH materials.
- Diffusion from the LiH surface to the interior of the material requires greater energy drive.
- The introduction of interstitial H atoms can make LiH have metal properties.
- The lowest thermal conductivity of the material can provide a reference for the selection of ceramic materials.
- LiOH is more thermally stable than LiH.

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ABSTRACT

LiH is valued by researchers and engineers for its high-temperature resistance, high strength, high hydrogen storage capacity of up to 12.6 wt%, and high neutron absorption cross section. However, sintering and application of this material are difficult. Lithium hydride is a highly effective neutron shielding material. The neutron shielding ability of lithium hydride is directly affected by the homogeneity of the total amount and distribution of hydrogen. The heat migration of hydrogen reduces the neutron shielding effectiveness of lithium hydride. It is critical to forecast H's diffusion behavior. The diffusion coefficient is a critical metric for determining the diffusion efficiency. As a result, the stability, electronic structure, and bonding properties of LiH, LiOH, and H entering the tetrahedral interstitial sites of LiH were calculated using first principles based on density functional theory (DFT). The elastic modulus of the system was also examined to find the lowest thermal conductivity, and some thermodynamic parameters were derived by calculating LiH phonons. Finally, static calculations were used to determine the diffusion coefficient and diffusion activation energy of H in the LiH solid solution. The results revealed that the addition of H altered the characteristics of the LiH semiconductor, giving it metallic features. The lowest thermal conductivity of the LiH system was 2.9. $W \cdot m^{-1} K^{-1}$. The diffusion coefficient was $D = (2.638 \times 10^{-2} \quad cm^2/s)\exp(-5.811 \times 10^{-3} \quad cal/K_BT)$. The activation energy for diffusion was 0.252 eV. The diffusion coefficients, elastic constants, and lattice parameters all corresponded well with previously published results. The density functional theory was employed to study the material's system energy and electrical

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structure, and the microscopic migratory behavior of hydrogen was predicted theoretically. The calculation of the lowest thermal conductivity provided a reference for the screening of ceramic materials.

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1. Introduction

Ceramic materials have been applied in various industries because of its high-temperature resistances, high strengths, optical transparency, and other properties. LiH has a high hydrogen atom density, a high melting point, and a low density. It has a greater neutron absorption cross section as well as a smaller molecular mass, making it a suitable neutron shielding material [1]. Solid-state hydrogen storage technology development is critical for the energy industry [2]. LiH may be utilized to store hydrogen as well. According to reports, LiH has a hydrogen storage capacity of up to 12.6 wt% [3,4]. The fundamental disadvantage of LiH is the severe conditions needed for hydrogen absorption and desorption. The dehydrogenation temperature must be 720 °C [5], and the hydrogenation temperature must be 700 °C or 50 MPa [6]. It should be emphasized, however, that the sintering of LiH is sluggish and complex. During sintering, it is expected that the densification process (mass transfer process) is improved while the non-densification process (surface mass transfer process) is inhibited. Sintering is a well-known irreversible thermodynamic process in which the system's free energy is lowered. The majority of the sintering process is a thermal activation process that relies on the ability of the atoms in the sintered body to receive sufficient energy to leave their present location and shift to a neighboring location. The activation energy of diffusion is an essential quantity. Heat and mass transfer are two of the most important scientific concerns in sintering.

Whether considering the LiH sintering process or its application as a neutron shielding material, the diffusion performance of H is crucial. H segregates and diffuses at flaws in materials, such as grain boundaries, causing anomalous expansion and even hydrogen embrittlement in metals and metal oxides [7–12]. The analysis of H in LiH is frequently calibrated based on the isotope of H in experimental approaches, and the diffusion coefficient of deuterium at a given temperature may be calculated using the thermogravimetric method [13]. The fundamental concept is to utilize Fick's diffusion equation to compute the deuterium diffusion coefficient in LiH by monitoring the change in the system weight. In the theoretical method, the diffusion coefficient of deuterium in LiH at different temperatures can be obtained by a thermodynamic model [14], and the predictions are identical to the experimental results.

The advancements in computer performance have brought great significance to materials science. Scholars have developed computational materials science through the use of computers to describe physical events at the atomic level and to calculate physical characteristics [15–21]. Density functional theory (DFT) [22] has been utilized by researchers to assess the energy and electrical characteristics of materials. Since 1930, there have been studies on LiH. The lattice constants and binding energies of LiH were computed by Hylleraas [22]. Banger et al. [23] found that the structure with 6H added in the LiH gap was the most stable, and with the addition of H, the LiH of the semiconductor obtained metal properties. By doping noble metals, Yu and Pan [24] altered the structural stability and dielectric characteristics of LiH. By using lithium-ion vacancy defects, Soufiane Bahou et al. [25] modified the hydrogen storage capacity. LiH not only hydrolyzes hydrogen [26–29] but also changes the surface of hydrogen storage materials [30–32]. Limiting LiH to high surface area graphite [33], nickel-coated LiH nanoparticles [34], and adding catalysts [35] can increase hydrogen storage performance and hydrogen absorption and desorption efficiency. However, the H diffusion mechanism in LiH is easily ignored.

In this study, the diffusion activation energy and diffusion coefficient were analyzed by first-principles. The relationship between the temperature and the diffusion process causes the system's heat transfer coefficient to be crucial. The calculations and measurements of the thermal conductivity of LiH have been mostly concentrated in the temperature range of 0–700 K [36–38]. However, the melting point of LiH is about 962 K. The lowest thermal conductivity has not been reported. The lowest thermal conductivity is helpful for the screening and application of ceramic materials [39]. In this study, the elastic constants and the lowest thermal conductivity of LiH were calculated by first-principles.

Based on the above analysis, the diffusion coefficient and the lowest thermal conductivity are critical characteristics for both LiH sintering and hydrogen storage. The lowest thermal conductivity of LiH ceramic materials has not been reported. When the temperature rises, the surface of LiH easily reacts with water to generate LiOH and H_2 [27]. During the sintering process, LiOH can react with LiH to generate Li_2O and H_2. In this study, first-principles calculations were used to investigate the stability, fundamental physical properties, electronic structures, and bonding features of LiH and LiOH, as well as the diffusion activation energy, diffusion coefficient, and minimum thermal conductivity of H in the LiH gap. The determination of this parameter contributes to a better understanding of the diffusion process of H in LiH, aids in the screening of ceramic materials, and serves as a reference for engineering practice.

2. Model establishment and calculation method

Based on density functional theory (DFT). The structures of LiH and LiOH were optimized in this study using the generalized gradient approximation (GGA) [40], the Cambridge

serial total energy package (CASTEP) calculation software [41], and the ultra-soft pseudopotentials [42]. Fig. 1 displays the results of the cut-off energy test, with the convergence accuracy set to 1.0 × 10^{-6} eV/atom and the K point set at [24] 12 × 12 × 12. The energy of the system typically became steady at about 650 eV. LiH belongs to the face-centered cubic crystal system at room temperature, and the space group number is Fm-3m (225). The optimized lattice constant of LiH was 4.008 Å, which is consistent with the calculated value of 4.009 Å [23] and the experimental value of 4.083 Å [43]. The LiOH space group number was *P4/nmm*, and the optimized lattice constants were a = b = 3.562 Å and c = 4.704 Å, which were consistent with the values of a = b = 3.487 Å and c = 4.279 Å calculated by other scholars [44].

The stability of crystals is evaluated by thermodynamics and kinetics [45–47]. The enthalpy of formation H_f can be used to represent the thermodynamic stability [48,49]. The enthalpy of formation represents the difficulty of the formation of a compound, and a negative value corresponding to released energy indicates that the compound is easy to form. The enthalpy formulas for the system under consideration are as follows:

$$H_f(Li_aH_b)=\frac{1}{a+b}\left[E_{tot}(Li_aH_b)-aE(Li)-\frac{b}{2}E(H_2)\right] \quad (1)$$

$$H_f(Li_aH_b+H)=\frac{1}{a+b+1}\left[E_{tot}(Li_aH_{b+1})-aE(Li)-\frac{b}{2}E(H_2)-\frac{1}{2}E(H_2)\right] \quad (2)$$

$$H_f(Li_aH_bO_c)=\frac{1}{a+b+c}\left[E_{tot}(Li_aH_bO_c)-aE(Li)-\frac{b}{2}E(H_2)-\frac{c}{2}E(O_2)\right] \quad (3)$$

Phonon dispersion is used to assess the solid's dynamic stability [50], The structure was calculated using the finite displacement approach in this study. We utilized the Arrhenius diffusion equation to calculate the diffusion coefficient of interstitial H atoms in the LiH system using a "2 × 2 × 2" supercell model, and the diffusion coefficient expression is as follows:

$$D=D_0\exp(-E_b/k_BT) \quad (4)$$

where D_0 is the exponential pre-factor, E_b is the diffusion energy barrier, and k_B is the Boltzmann constant. According to the relationship between the jump frequency and the diffusion coefficient [51], the relationship between the diffusion coefficient and temperature can be obtained:

$$D=l^2\nu\exp(-E_b/k_BT) \quad (5)$$

The jump frequency can be calculated by the following formula [52,53]:

$$\nu=\sqrt{E_b/2ml^2} \quad (6)$$

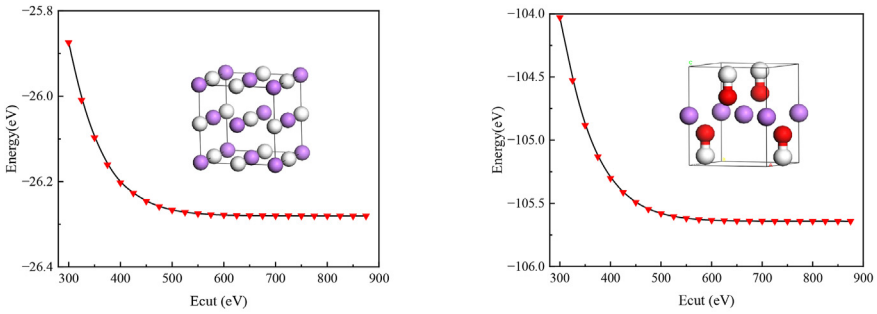
where $m_H = 1.67\times10^{-27}$kg, l is the transition distance of the hydrogen atom (l = 2.006 Å), and E_b is the diffusion activation energy. The diffusion activation energy can be calculated using the function of the transition state in CASTEP. The CASTEP transition state algorithm employs the "Complete LST/QST" transition state search approach. This approach combines the standard LST/QST algorithm with the conjugate gradient method, which may rapidly and reliably determine the appropriate transition state [54]. The approach described above is a static computation of the diffusion coefficient of H in an interstitial solid solution. The dynamic approach to determine the diffusion coefficient can be introduced by the Einstein equation. The diffusion coefficient was estimated using Einstein's mean square displacement formula (7).

$$D=\frac{1}{N}\sum_{i=1}^{N}\frac{\langle r_i(t)-r_i(0)\rangle^2}{2dt}=\frac{R^2}{2dt} \quad (7)$$

where R^2 is the mean square displacement, and d is the dimension (d = 3).

The overall tendency of the thermal conductivity change in a ceramic material is that at extremely low temperatures, the thermal conductivity is determined by the material's physical size, grain size, and dislocation distance. In this temperature range, the thermal conductivity increases fast with temperature and typically exponentially. When the thermal conductivity hits its maximum, it begins to fall. In this temperature range, the thermal conductivity is dictated by phonon vibration collisions and is extremely sensitive to the defect concentration. When the temperature is near the melting point of the crystal, the thermal conductivity is unaffected by the temperature and is close to the material's lowest thermal conductivity. The following are the formulas for the thermal conductivity [39]:

$$\kappa_{min}=\kappa_B\upsilon_m\lambda_{min} \quad (8)$$



(a). LiH cut-off energy test diagram (b). LiOH cut-off energy test graph

Fig. 1 – Crystal structure cutoff energy test.

Table 1 – Lattice constants, volumes, and formation enthalpies of LiH, LiOH, and LiH + H (eV/atom).

Crystals structure	a = b (Å)	c (Å)	V (Å^3)	H_f (eV/atom)
LiH	4.008	4.008	64.38	−1.365
LiOH	3.487	4.279	52.03	−1.751
LiH + H	4.157	4.157	71.84	−1.305

$$\kappa_{min} = 0.87\kappa_B N_A^{2/3} \frac{n^{2/3}\rho^{1/6}E^{1/2}}{M^{2/3}} \tag{9}$$

where κ_B is Boltzmann's constant, N_A is Avogadro's constant, λ_{min} is the average free path of the smallest phonon, n is the number of atoms in each primitive cell, M is the mass of the molecule in the protocell, ρ is the cell density, and E is Young's modulus.

The equations of the shear modulus G_V given by Voigt [55] and the elastic constant G_R given by Reuss [56] are respectively as follows:

$$G_V = \frac{C_{11} - C_{12} + 3C_{44}}{5} \tag{10}$$

$$G_R = \frac{5(C_{11} - C_{12})C_{44}}{4C_{44} + 3(C_{11} - C_{12})} \tag{11}$$

Hill [57] demonstrated that the two approximations characterized the polycrystalline's top and lower boundaries, respectively, and that the true shear modulus was the arithmetic average of the two. That is, the following formulas may respectively be used to calculate the shear modulus G and the bulk modulus B:

$$G = \frac{G_V + G_R}{2} \tag{12}$$

$$B = \frac{C_{11} + 2C_{12}}{3} \tag{13}$$

Young's modulus E and Poisson's ratio μ are calculated respectively as follows:

$$E = \frac{9GB}{3B + G} \tag{14}$$

$$\mu = \frac{3B - 2G}{2(3B + G)} \tag{15}$$

The destruction of intermolecular chemical bonds when creating the surface of the material is reflected by the surface energy. The calculation formula is as follows:

$$E_{surf} = (E_{slab} - N \cdot E_{bulk}) / 2A_{surf} \tag{16}$$

E_{slab} is the total energy of the system after the optimization of the cut surface, E_{bulk} is the energy of the LiH crystal, N is the number of LiH crystals contained in the surface system and A_{surf} is the surface area.

The stability of the adsorbate can be measured by adsorption energy. Where n takes 1 or 2.

$$E_{abs-H_n} = E_{slab+H_n} - E_{slab} - \frac{1}{(3-n)}E_{H_2} \tag{17}$$

The surface roughness is described by the interlayer relaxation rate Δdmn, which d_0 represents the initial interlayer spacing and dmn represents the distance from the first layer to the first layer.

$$\Delta dmn = (dmn - d_0) / d_0 \times 100\% \tag{18}$$

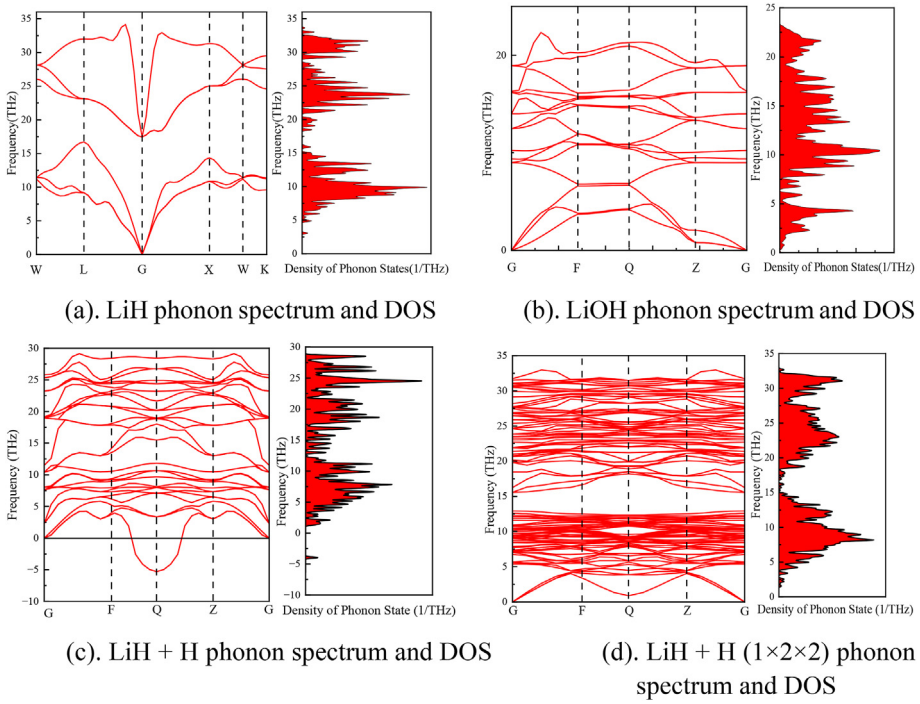


Fig. 2 – Kinetic stability of crystal structure.

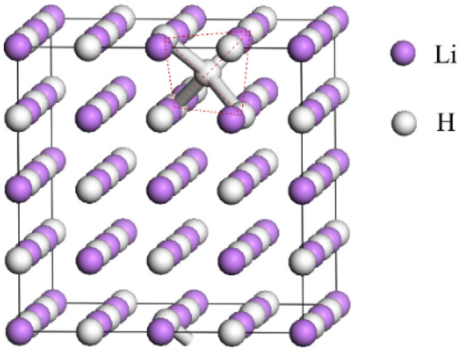


Fig. 3 – LiH + H tetrahedral interstitial position.

3. Results and discussion

3.1. Crystal structure stability analysis

First, it is necessary to analyze the thermodynamic stability and kinetic stability of the structure before studying other properties. The thermodynamic stability is determined by the formation enthalpy. The lattice constant, volume, and formation enthalpy after structural optimization are shown in Table 1.

As shown in Table 1, the thermodynamic stability of the three crystal formations was guaranteed. For a negative enthalpy of formation, it is generally understood that the more negative the enthalpy of formation is, the more stable the thermodynamics of the structure is. Of the crystal structures examined, LiOH was the most thermodynamically stable, indicating that as the temperature rose, LiH would easily react with H_2O to create LiOH and H_2 [27]. The introduction of H into the LiH interstitial sites promoted lattice expansion, reducing the structure's thermal stability. The dynamic stability of the material was determined by the atom's vibrational frequency, and the phonons represent the quantization properties of the lattice vibration energy [58]. The stability of the lithium hydride dynamics was evaluated by the phonon scattering curve and the structure's phonon state density, as illustrated in Fig. 2, to further emphasize the structure's stability. If there are no imaginary frequencies in the phonon spectrum, the structural dynamics are stable. Fig. 2(c) shows that there were imaginary frequencies at the high symmetry point Q, indicating that H was dynamically unstable in the unit cell LiH, so it expanded. After expansion, the system dynamics were stable, as shown in Fig. 2 (d). The phonon spectrum included a total of 3n branches, comprising 3 acoustic branches and 3n−3 optical branches for the material unit cell containing n atoms. The optical branch indicated that there were relative vibrations between the atoms in the cell, while the acoustic branch reflected the cell's total vibrations.

3.2. Electronic structure and bonding properties

To learn more about the system's electrical structure as well as the H occupancy and bonding properties in the LiH system, we began by calculating the structure's energy band diagram,

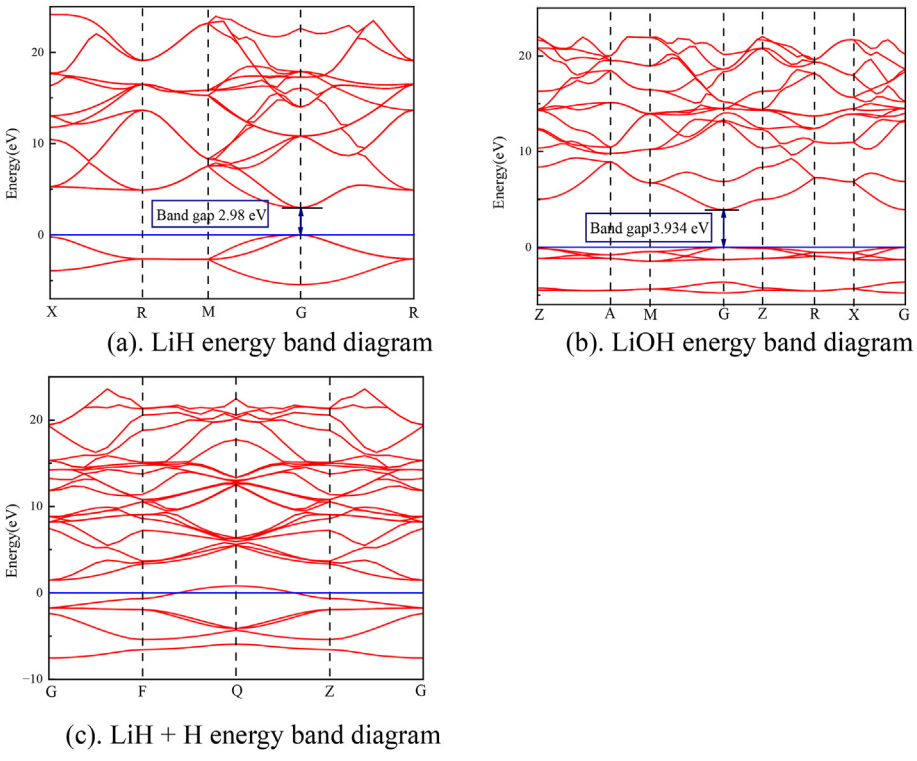


Fig. 4 – Crystal structure energy band analysis.

Table 2 – Analysis of the atomic Mulliken layouts of the three systems.

Crystals structure	Charge spillage parameters	Element	Net charge			Mulliken charge (e)	Hirshfeld charge (e)
			s	p	Total		
Li_4H_4	0.06%	Li_4	1.69	0.91	2.60	0.40	0.13
		H_4	1.40	0.00	1.40	−0.40	−0.13
$Li_2O_2H_2$	0.37%	Li_2	2.08	0.35	2.43	0.57	0.22
		H_2	0.67	0.00	0.67	0.33	0.15
		O_2	1.85	5.05	6.90	−0.90	−0.36
$Li_{32}H_{32} + H$	0.06%	Li_{28}	1.70	0.91	2.61	0.39	0.13
		Li_4	1.69	0.87	2.56	0.44	0.14
		H_{32}	1.39	0.00	1.39	−0.39	−0.13
		H^{int}	1.27	0.00	1.27	−0.27	−0.10

which is shown in Fig. 4(a) and (b). The computed band gap for LiH was 2.98 eV, which was quite compatible with the band gap of 3.002 eV and the band diagram for LiH reported previously [24]. The LiOH band gap was estimated to be 3.934 eV. As shown in Fig. 3, H was located in the tetrahedral interstitial location of the LiH system. As shown in Fig. 4(c), the addition of H caused the entire system to exhibit metallicity.

The Mulliken charge reflects the local charges of atoms. By performing the Mulliken layout analysis of the three systems, the bonding and charge distribution of the system were studied, and the results are shown in Table 2. The electron gain and loss in an atom were measured by the number of charges, which was related to the bonding characteristics of the atom. According to Table 2, the charge distributions of the same atom in the LiH and LiOH systems were consistent because they were uniformly arranged and the system was electrically neutral. However, the entry of H into the LiH interstitial site broke this balance and changed the charge distribution, resulting in the Mulliken charge of the system not being zero. This also showed that the addition of H caused LiH to exhibit metallicity. From the Mulliken layout, it was calculated that the bond length was less than 3 Å. In Table 2, H^{int} is an interstitial H atom. The Hirshfeld charge is a description of the affinity of the charge within an atom or molecule. H served as an electron acceptor in the LiH system, whereas Li served as an electron donor. Table 2 shows that the charge transmission and reception were both equivalent. The tetrahedral interstitial site was occupied and bound to the surrounding atoms by the addition of H atoms, which modified the charge distribution and created free electrons during the charge transfer process. From Fig. 5, which shows the charge difference density diagram, it can also be seen that the transferred amounts for the interstitial H atoms and surrounding atoms decreased. The Mulliken charge shown in Table 2 also illustrates this point. In the LiOH system, the H atom transformed into a charge donor under the effect of O.

3.3. *Calculation of thermodynamic parameters*

According to the phonon calculations of the LiH primitive cell, the zero point energy of the LiH system was 0.226 eV. The calculation results of the enthalpy, free energy, and entropy of LiH are shown in Fig. 6. The entropy was expressed by temperature T × entropy to compare with the enthalpy. Fig. 7 displays the variations of the Debye temperature and heat capacity of the system with the temperature.

The elastic stiffness tensor C_{ij} of the LiH calculated by CASTEP had three independent components: C_{11}, C_{44}, and C_{12}. The calculated data were compared with existing literature values, and the results were in good agreement. The elastic constants were calculated using Formulas (10)–(14) and are shown in Table 3. The mechanical stability of the crystal could be determined by $C_{11} + 2C_{12} > 0, C_{44} > 0, C_{11} - C_{12} > 0$. The bulk modulus B characterizes the ability of the material to resist fracture. The shear modulus G measures the plastic deformation. B/G can be used as a measure of ductility or brittleness. The calculation for isotropic cubic crystals yielded B/G = 1.67 as the critical value of the ductility–brittleness ratio of materials [59]. A B/G value greater than the critical value indicates high ductility, and a B/G value less than the critical value indicates high brittleness. For LiH, the B/G value was 0.833, indicating that the compound had a certain brittleness.

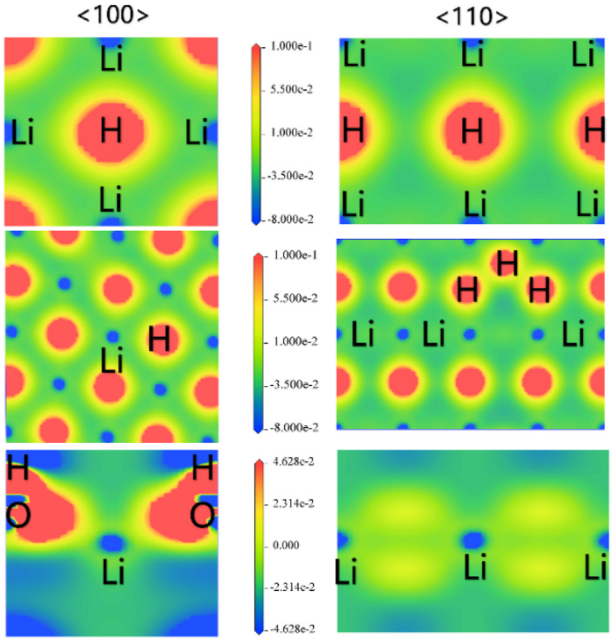


Fig. 5 – Surface atomic charge difference density diagrams of three systems (from top to bottom: LiH, LiH + H, and LiOH).

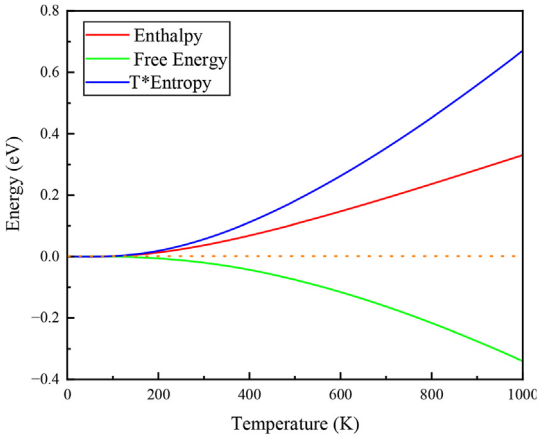


Fig. 6 – Enthalpy, free energy, and entropy variations of LiH with temperature.

At the same time, the average speed of sound [60] was calculated to be 7831.46432 (m/s), and the elastic Debye temperature was 1163.60823 K. According to Formula (9), the lowest thermal conductivity of LiH was 2.9 W m^{-1} K^{-1}.

3.4. *Diffusion activation energy and diffusion coefficient*

3.4.1. *H adsorption and diffusion on LiH(001)surface*

The LiH(001) surface model was established as shown in Fig. 8. Through the surface energy calculation formula (16), the surface energy of the 4 layers and the 6 layers was calculated. The results show that the surface energy of the 6 layers is only 0.38% lower than that of the 4 layers. This shows that 4 layers can meet the calculation requirements. After optimizing the surface structure of LiH(001), the interlayer relaxation rate is calculated by formula(18). Li collapses 2.74% inward and H expands 0.05% outward.

The adsorption energy may be used to assess the stability of H adsorption on various LiH (001) surface locations. Formula (17) may be used to compute the adsorption energy. H adsorbed on the Li top site has an adsorption energy of −0.68 eV, while H adsorbed on the H-H bridge site has an adsorption energy of −0.9 eV. As a result, H adsorbed on the LiH (001) surface's H-H bridge site is more stable. The diffusion activation energy of H from the LiH (001) surface to the bulk is calculated to be 0.67 eV using the transition state.

3.4.2. *Interstitial H diffusion in LiH*

First, H was located in the tetrahedral sites of LiH. The diffusion of H atoms induced by the concentration gradient was not taken into account in this work, only the diffusion of H atoms produced by thermal motion was. The "2 × 2 × 2" supercell model was used for the LiH. The diffusion of H compressed the surrounding atoms, causing them to move, and the addition of H deformed the lattice. As shown in Fig. 9, we supposed that H diffused to nearby interstitial sites in the (001) direction. The reactants and products with the lowest energy after optimization were set up, and the diffusion activation energy of H at the LiH interstitial position was calculated by using the "Complete LST/QST" algorithm. The diffusion activation energy was 0.252 eV, and the results are shown in Fig. 10. The activation energy of the self-diffusion coefficient in Ref. [13] is 1.054 eV. The diffusion barrier of H in the LiH interstitial solid solution was not large, so H could diffuse more easily. The transition state confirmation is given in Fig. 11 to demonstrate the validity of the obtained activation energy. We compute the diffusion activation energy of H along the three directions (100), (010), and (110) using the approach described above. The results reveal that the diffusion activation energy in the directions (100) and (010) is consistent with that in the direction (001), which is 0.25 eV (two decimal places are kept). The (110) direction has a value of 1.1 eV. As a result, it is appropriate to adopt the (001) diffusion path for H.

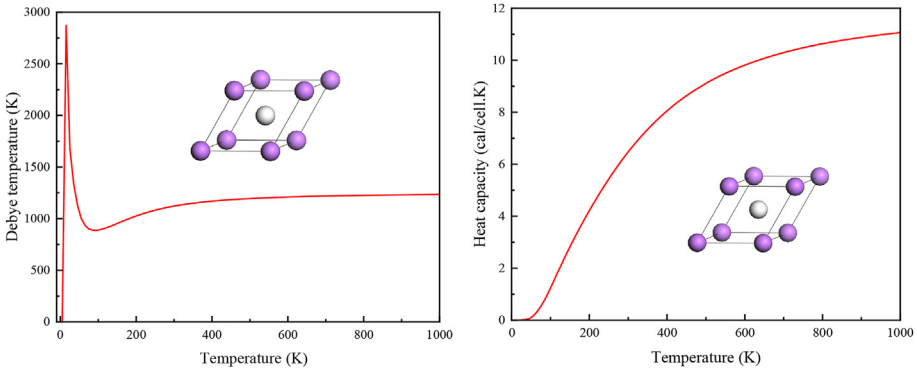


Fig. 7 – LiH Debye temperature (left) and heat capacity (right) variations with temperature.

Table 3 – Elastic constants (GPa).

C_{11}	C_{44}	C_{12}	G_V	G_R	G	B	E
82.021	48.6157	12.1869	43.1363	42.0215	42.5789	35.4650	91.228
80.0 [61]	50.65 [61]	12.83 [61]					
77.0 [62]	48.2 [62]	12.7 [62]					

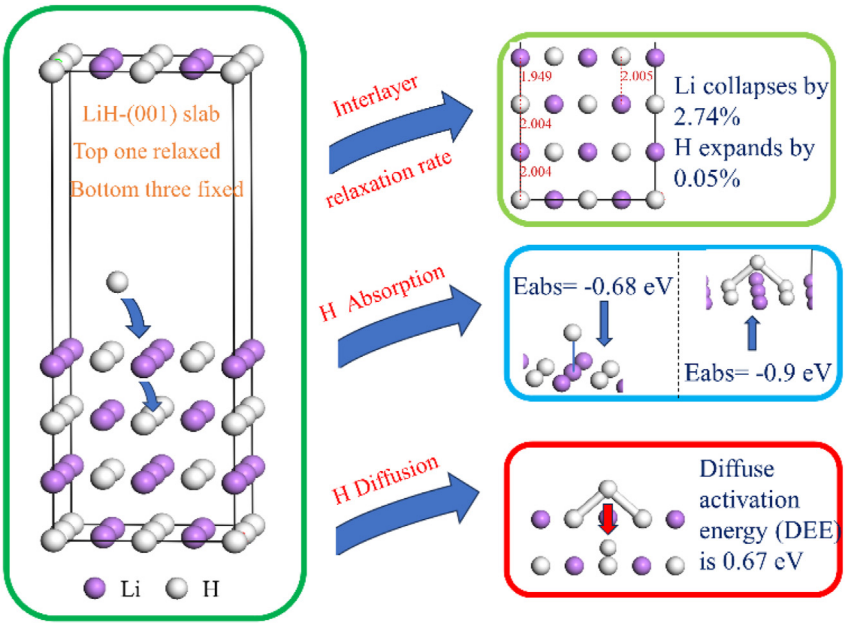


Fig. 8 – Theoretical calculation process diagram (interlayer relaxation rate, H adsorption energy, H diffusion activation energy).

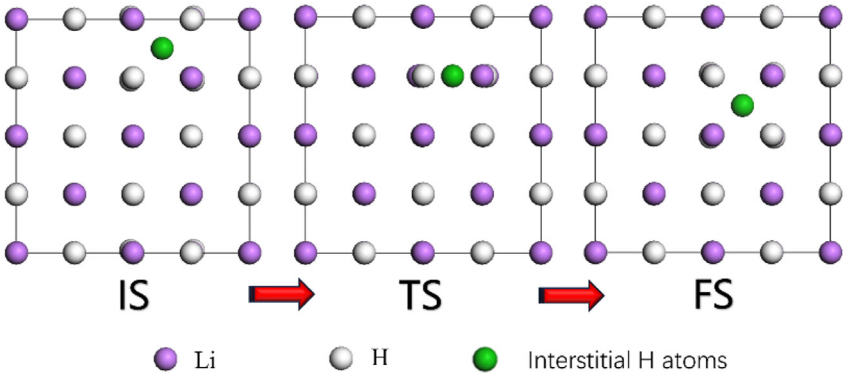


Fig. 9 – Diffusion path of H in LiH (side view).

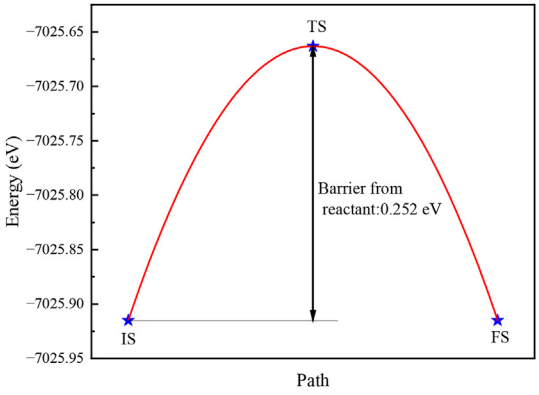


Fig. 10 – H diffusion activation energy diagram.

Through Formulas (4)–(6), the diffusion coefficient expression of H in the LiH interstitial solid solution can be obtained as follows:

$$D = (2.638 \times 10^{-2} \quad cm^2/s)\exp(-5.811 \times 10^{-3} \quad \text{cal} \ / K_B T).$$

The results calculated in a previous study [13] were as follows:

$$D = (2.41 \times 10^{-2} \quad cm^2/s)\exp(-24.3 \times 10^{-3} \quad \text{cal} \ / K_B T).$$

The computed values differed from the self-diffusion coefficients found in Refs. [13,63]. The fundamental reason for this was that the theoretical model for interstitial hydrogen ion diffusion had a larger vacancy concentration than the self-diffusion process. The computed diffusion coefficient exceeded the self-diffusion coefficient. The diffusion coefficient

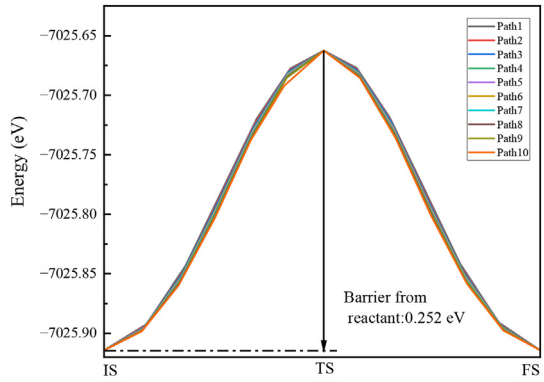


Fig. 11 – H diffusion transition state confirmation.

increased dramatically as the temperature increased. The diffusion rate of H in the LiH interstitial solid solution was relatively fast.

4. Conclusions

To summarize, we used first-principles to compute the energy and electronic structure of LiH, LiOH, and LiH + H systems. The mechanical and thermodynamic properties of LiH, as well as the adsorption and diffusion mechanisms of H on the surface and interior of LiH, were investigated. The following are the primary conclusions:

1. The thermal stability of LiOH is more stable than that of LiH.
2. The introduction of interstitial H changes the charge distribution and produces free electrons, which makes LiH have metal properties. At the same time, the elastic constant of LiH was calculated to be $C_{11} = 82.02$, $C_{44} = 48.62$, and $C_{12} = 12.19$. We also calculated the minimum thermal conductivity (2.9 $W \cdot m-1 \cdot K-1$) of LiH, which is crucial for the screening of ceramic materials.
3. The H-H bridge site is the stable adsorption site of H on the LiH (001) surface. Inside LiH, H occupies the tetrahedral interstitial position. The diffusion activation energy of H is 0.67 eV from the surface to the inside of the body, and it is 0.252 eV in the internal gap of the body. Because, after cutting the surface, the surface atoms will expand or contract, causing the system's electron distribution to alter and resulting in a greater diffusion energy barrier.

Data availability statement

The data that support the findings of this study are available from the corresponding author, Huayan Chen, upon reasonable request.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests.

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